

CONVOCATORIA

“Investigación Humanística 2025”

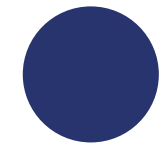


PROYECTO DE INVESTIGACIÓN

Teoría y aplicaciones de control óptimo estocástico.

*Facultad de Ciencias en Física y
Matemáticas.*





Responsable técnico

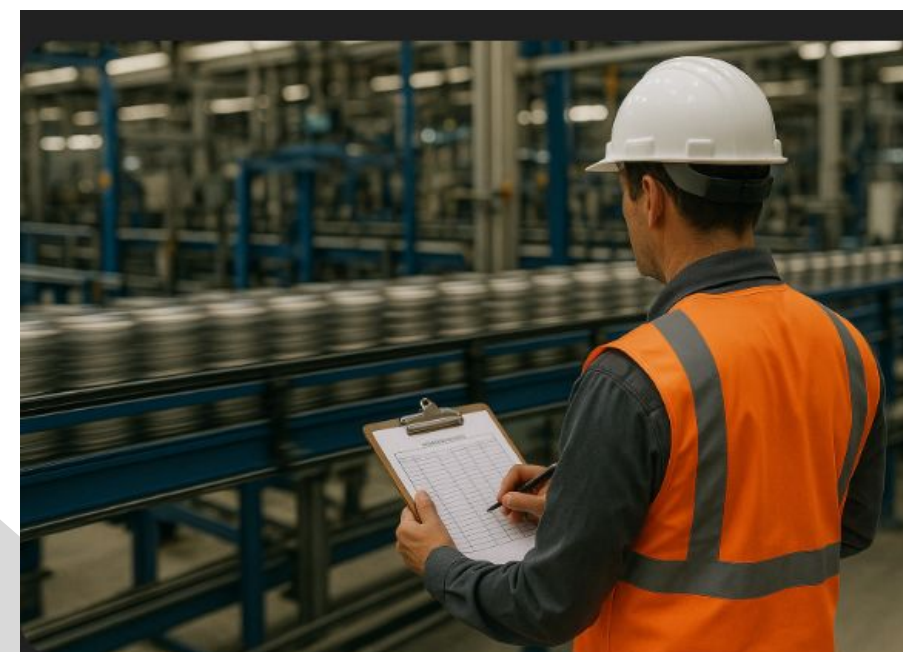
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ESPACIO DE INCIDENCIA

- Salud pública
- Monitoreo de transporte (sistemas dinámicos parcialmente observables)



TRABAJOS PREVIOS

<https://jennlg.quarto.pub/tesis-jslg/>

<https://jazielmartinezl.quarto.pub/tesis-1/>

<https://mara-ld.quarto.pub/httpsmara-ldquartopubtesis-mdl/>

<https://quartopub.com/sites/01eb0b99-8be3-4afa-8e51-3c6dc5d90ff3>

<https://damian-esco.quarto.pub/bio-heat-with-deeponet/>

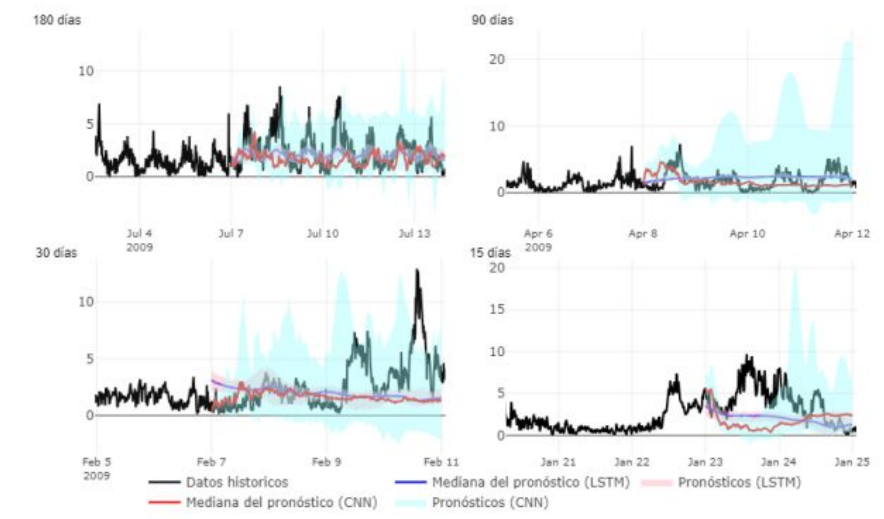
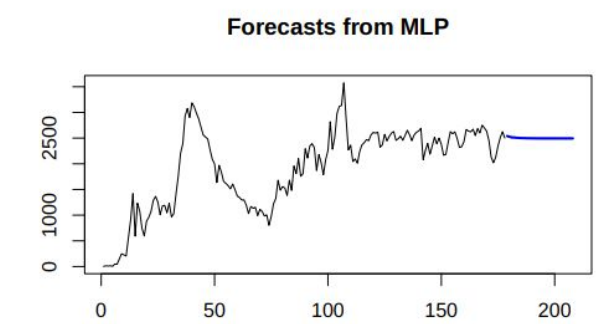
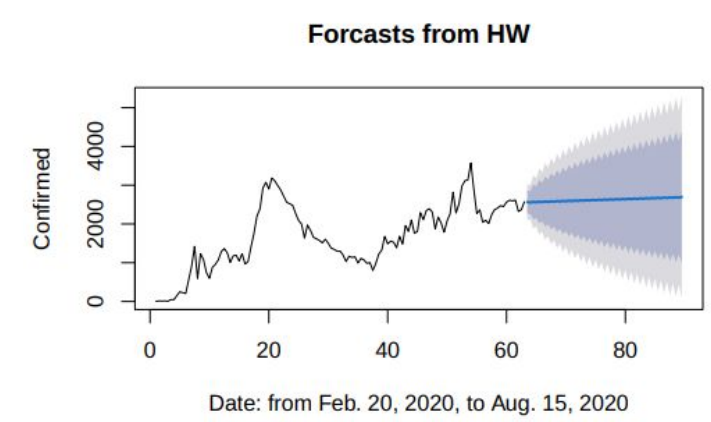
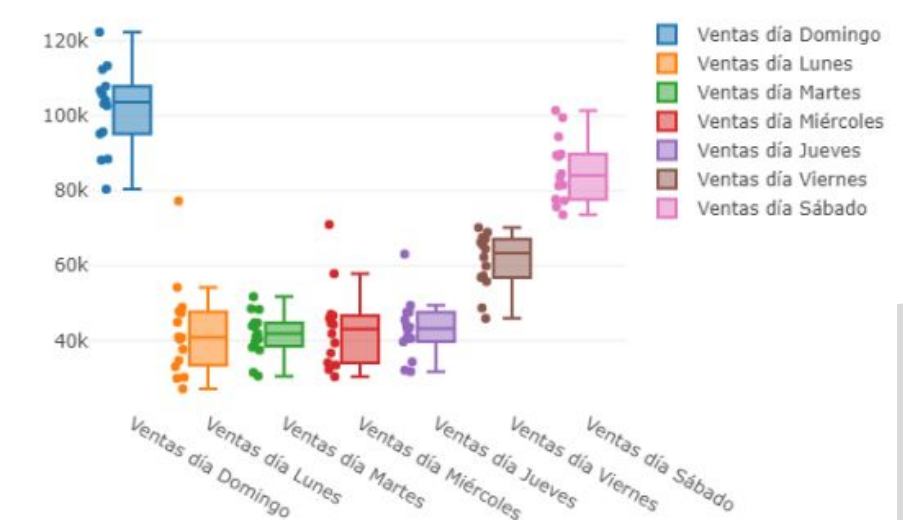


Figura 11.17.: Predicción de 7 días, 4 días y 2 días de la velocidad del viento utilizando la red neuronal convolucional y la red neuronal LSTM

$$\rho c \frac{\partial T}{\partial t} = k_{\text{eff}} \frac{\partial^2 T}{\partial x^2} - \rho_b c_b \omega_b (T - T_a) + Q, \quad x \in \Omega, t \in [0, t_f] \tag{9.1}$$

Tabla 9.1.: Tabla de nomenclaturas de la Ecuación 9.1.

Símbolo	Descripción	Unidades
T	Temperatura del tejido	$^{\circ}\text{C}$
ρ	Densidad del tejido	$\frac{\text{kg}}{\text{m}^3}$
c	Calor específico del tejido	$\frac{\text{kg}^{\circ}\text{C}}{\text{W}}$
k_{eff}	Conductividad térmica	$\frac{\text{m}^{\circ}\text{C}}{\text{kg}}$
ρ_b	Densidad de la sangre	$\frac{\text{kg}}{\text{m}^3}$
c_b	Calor específico de la sangre	$\frac{\text{J}}{\text{kg}^{\circ}\text{C}}$
ω_b	Tasa de perfusión sanguínea	$1/\text{s}$
T_a	Temperatura arterial	$^{\circ}\text{C}$
$Q = q_m + q_p$	Fuente de calor	$\frac{\text{W}}{\text{m}^3}$
q_m	Metabolismo	$\frac{\text{W}}{\text{m}^3}$
q_p	Externa	$\frac{\text{W}}{\text{m}^3}$



AVANCES



An integrated mathematical epidemiology and inventory model for high demand and limited supplies under uncertainty

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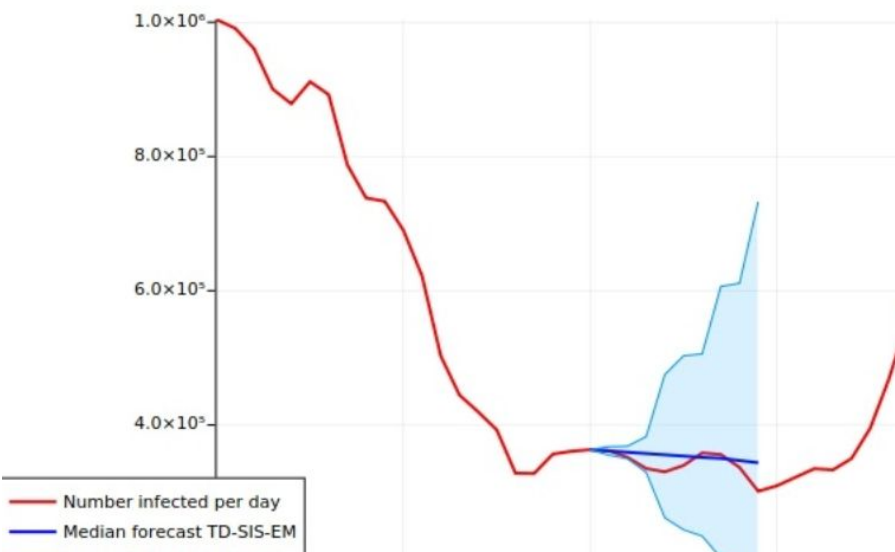
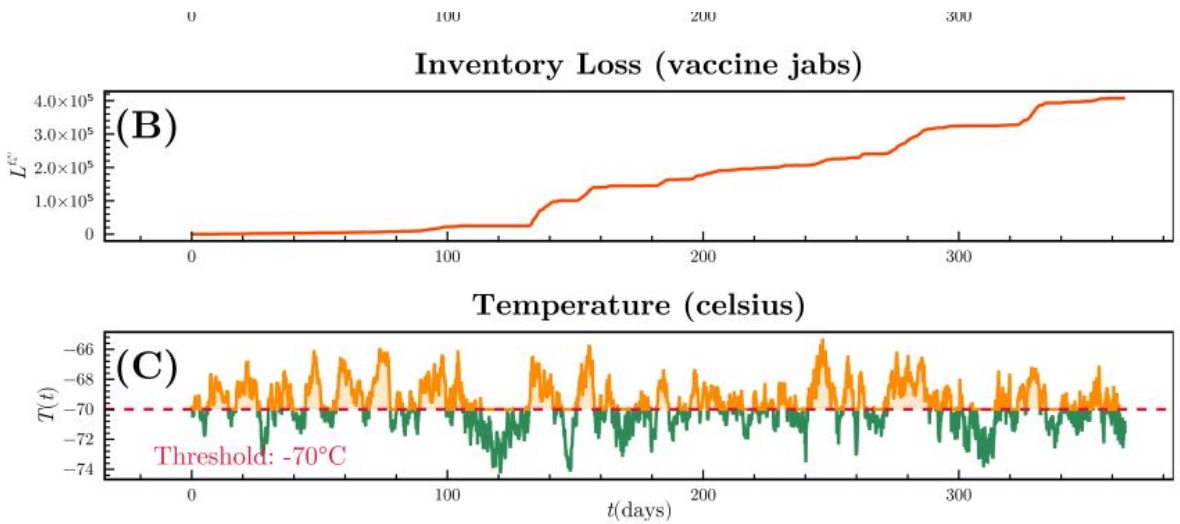
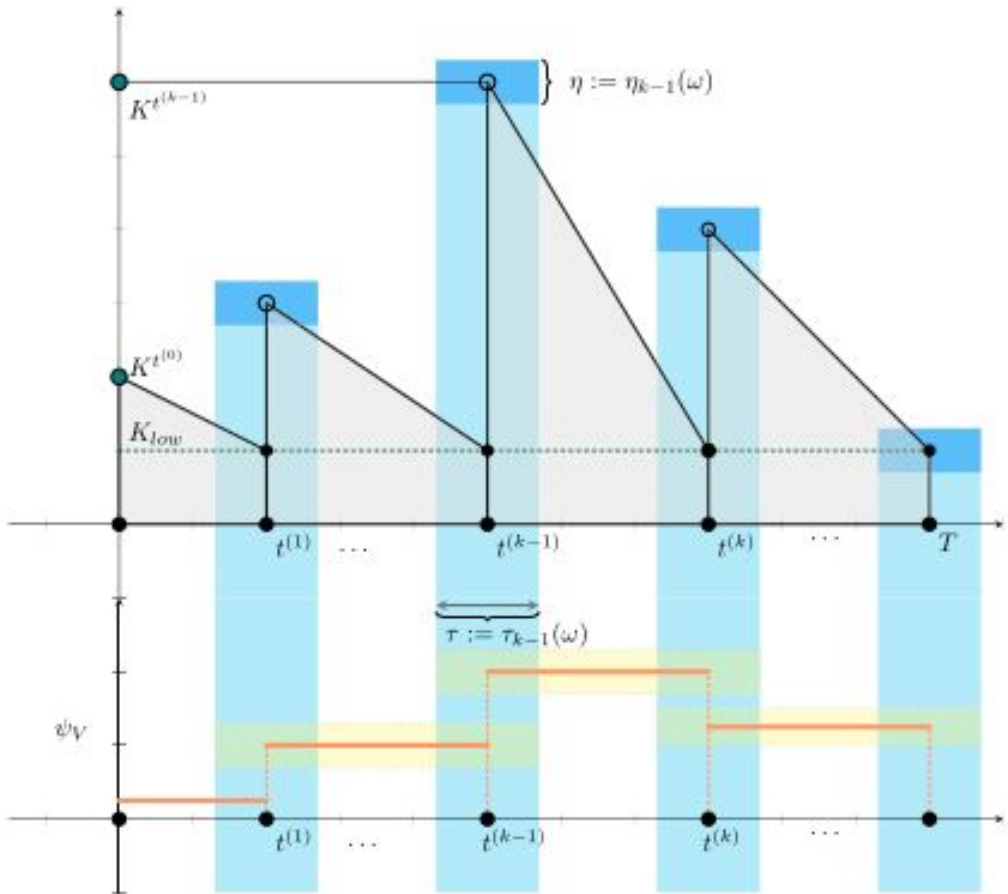
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ARTICLE INFO

Keywords:
 Mathematical epidemiology
 Vaccine inventory management
 Sequential decisions problem
 Markov decision process
 Simulation

ABSTRACT

At the start of the Coronavirus Disease (COVID-19) vaccination campaign in Mexico, the vaccine was the world's most essential and scarce asset. Managing its administration to optimize its use was, and still is, of paramount importance. However, when the first vaccine was developed at the end of 2020, due to unprecedented demands and early manufacturing of vaccines, decision-makers had to consider the management of this asset with high uncertainty. We aim to analyze how random fluctuations in reorder points and delivery quantity impact the mitigation of a given outbreak. Because decision-makers would need to understand the implications of planning with a volatile vaccine supply, we have focused our effort on developing numerical tools to evaluate vaccination policies. One of our main objectives is to determine how many vaccines to administer per day so that a hypothetical vaccine inventory keeps its integrity while optimizing the mitigation of the outbreak. Our research uses classic models from inventory management and mathematical epidemiology to quantify uncertainty in a hypothetical vaccine inventory. By plugging a classic inventory model into an epidemic compartmental structure, we formulate a problem of sequential decisions. Then, we investigate how the random fluctuations in the reorder time and number of doses in each delivery impact a hypothetical ongoing vaccine campaign. Our simulations suggest that sometimes, it is better to delay vaccination until the vaccine supply is large enough to achieve a significant response.



$$dl_t = I_t \left([\beta(t)(1-I_t) - \gamma] dt + \sigma(t)(1-I_t) dW_t \right), \quad \beta(t) \in [\underline{\beta}, \bar{\beta}], \quad \sigma(t) \in [\underline{\sigma}, \bar{\sigma}].$$